Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

Inventor Information

of inventors: 1

1. **Eric Haywood**

Residence Information
US Residency
Fort Collins, CO
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Application Information

Customer number	**221639 -**
Correspondence address	---
Title of invention	**PREDICTIVE CROSS-VECTOR THREAT MODELING SYSTEM WITH TIME-REVERSAL NETWORK FORENSICS**
Attorney docket number	---
Entity status	---

Application type	Provisional Application under 35 USC 111(b)
Subject matter	Utility
Total number of drawing sheets	---
Filing by reference	No

Representative Information

of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

Domestic Benefit/National Stage Information

of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

Foreign Priority Information

of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Data was not provided for this section.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE:
This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. **Priority Document Exchange (PDX)**
Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the China National Intellectual Property Administration (CNIPA), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its

related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h) (1).

B. **Search Results from U.S. Application to EPO**

Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

NOTE:

Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Applicant Information

of applicants: 0

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

Data was not provided for this section.

Assignee Information including Non-Applicant Assignee Information

of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

HeadySystems Inc.
149 Remington Street Unit 425
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Organization

Signature

NOTE:

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

Signature	First name	Last name	Registration #	Date
/Eric Haywood/	ERIC	HAYWOOD	---	03/03/2026

UNITED STATES
PATENT AND TRADEMARK OFFICE

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
63/995,278	**03/03/2026 01:35:43 PM Z ET**	**-**

Title of Invention

PREDICTIVE CROSS-VECTOR THREAT MODELING SYSTEM WITH TIME-REVERSAL NETWORK FORENSICS

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	3518	FILED BY	ERIC HAYWOOD
PATENT CENTER #	74696231	FILING DATE	-
CUSTOMER #	221639	FIRST NAMED INVENTOR	Eric Haywood
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
generatedADS74696231.pdf	5	Application Data Sheet	109 KB
HS-2026-005-Final-Assigned-Updated-SPEC.docx	3	Specification	39 KB

Warning: One or more pages are missing page numbering. Page numbering will be automatically applied after submission.

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
generatedADS74696231.pdf	7386EF8048829CE336ABFF6F12A29D79503C148EEE0415CE3

7D9D8D5DE7117B46586E328FF174CE03D9068F856F7627C0E
D431824227C917394BC9474E3603B7

HS-2026-005-Final-Assigned-Updated-SPEC.docx

D3668F11200C41E0651E895F1CF6A6624FDE7CAF6F7B31307
EFB31D1567236D9E67C052E5940B23ACEAF88FE4199293CBE
CAE944B02A8A122524FF5A6D3E6D2A

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

PREDICTIVE CROSS-VECTOR THREAT MODELING SYSTEM WITH TIME-REVERSAL NETWORK FORENSICS

Title of the Invention	Predictive Cross-Vector Threat Modeling System with Time-Reversal Network Forensics
Docket Number	HS-2026-005
Inventor(s)	Eric Haywood
Residence(s)	149 Remington Street Unit 425, Fort Collins, CO 80524
Correspondence Address	HeadySystems Inc., 149 Remington Street Unit 425, Fort Collins, CO 80524
Attorney/Agent (Name & Reg. No.)	None
U.S. Government Interest (Yes/No)	No
Gov. Agency & Contract No.	N/A

Title of the Invention

[0001] Predictive Cross-Vector Threat Modeling System with Time-Reversal Network Forensics

Cross-Reference to Related Applications

[0002] This application is related to co-pending provisional applications Docket Nos. HS-2026-001 through HS-2026-004, filed concurrently herewith by the same inventor and assignee.

Field of the Invention

[0003] The invention relates to cybersecurity and intrusion detection, and more particularly to a predictive threat modelling system that detects attacks before hard thresholds are breached by analysing statistical deviations from baseline operational states and tracing attack origins through network routing table reversal.

Background of the Invention

[0004] Traditional intrusion detection systems rely on fixed thresholds for detecting SYN floods, ping floods and brute-force attempts. When network traffic exceeds a static threshold, an alert is triggered. This reactive approach allows attacks to proceed and cannot adapt to changing baseline conditions. Moreover, fixed-threshold systems do not trace attack origins through routing infrastructure, leaving lateral movement unchecked.

Summary of the Invention

[0005] The invention provides a predictive threat modelling system comprising four components: (1) a baseline modeler using machine learning to map normal operational states from container log analysis; (2) a dynamic threshold adjuster that computes detection thresholds as baseline metrics multiplied by $\varphi \approx 1.618$; (3) a pre-emptive action engine that detects sustained statistical deviations and initiates protective measures; and (4) a time-reversal path tracer that reverses network routes to identify origin nodes.

[0006] Dynamic thresholds are determined by matching metrics against baseline patterns. If metric data follows a baseline pattern, the baseline becomes the dynamic threshold after sanity checks; otherwise, a composite of baselines is used【160932740416553†L424-L486】【160932740416553†L531-L540】. This automatic determination replaces manual threshold tuning and adapts to changing conditions.

[0007] Time-reversal forensics extend worm origin identification and propagation path reconstruction techniques used in digital forensics. Algorithms like Origins identify worm origins and reconstruct propagation paths by tracing connections backward in time【88695340627218†L14-L37】【88695340627218†L69-L74】. The invention applies a similar technique to network attack payloads, traversing routing tables in reverse order to isolate compromised nodes.

Detailed Description of the Preferred Embodiment

A. Baseline Modelling

[0008] The system ingests container logs from services such as etcd, kube-apiserver and systemd to build a multidimensional baseline model capturing temporal patterns, resource utilisation and network traffic composition.

B. Dynamic Thresholding

[0009] Detection thresholds are computed dynamically as φ × baseline, e.g., SYN_THRESHOLD = IPTrie.root.volume × φ. Heavy hitters are detected when an IP address exceeds φ × the total traffic volume.

C. Pre-Emptive Action

[0010] Sustained positive acceleration toward a threshold triggers pre-emptive measures: rerouting traffic, throttling suspicious ranges and increasing logging granularity. The system calculates time-to-threshold breach and acts before a hard threshold is crossed.

D. Time-Reversal Network Forensics

[0011] Upon detecting a malicious payload, the system captures the routing path and traverses it in reverse, consulting network routing tables (e.g., in etcd) to identify the origin node or compromised pod. This process adapts digital forensics algorithms for worm propagation path reconstruction[2].

References

1. [1] Dynamic thresholding improves intrusion detection by matching metric patterns against baselines; if a matching pattern is found, its baseline becomes the dynamic threshold after sanity checks, otherwise a composite baseline is used【160932740416553†L424-L486】【160932740416553†L531-L540】.
2. [2] Worm origin identification and propagation path reconstruction are essential for digital forensics; algorithms like Origins identify worm origins and reconstruct propagation paths by tracing connections backward in time【88695340627218†L14-L37】【88695340627218†L69-L74】.